Lecture 29 Factor Analysis I: Model and Estimation

STAT 441: Statistical Methods for Learning and Data Mining

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Outline

1. Motivation
2. Factor Model
3. Principal Component and Maximum Likelihood Solution
4. Summary and Remark

Motivation

- Suppose that variables are grouped, with those in the same group being highly correlated with each other and largely uncorrelated with those in other groups.
- We might try to explain the within-group correlations in terms of a small number of unobserved, random factors.
- Example: highly correlated scores in academic subjects, and highly correlated performances in athletic activities, might be explained by two factors — intelligence and athletic ability.
- The choices made here are often controversial (might social or cultural factors account just as well for the groupings?).

Definition and Purposes of FA

Factor Analysis (FA) assumes the covariation structure among a set of variables can be described via a linear combination of unobservable (latent) variables called factors.

There are three typical purposes of FA:

- Data reduction: explain covariation between p variables using $r < p$ latent factors
- Data interpretation: find features (i.e., factors) that are important for explaining covariation (exploratory FA)
- Theory testing: determine if hypothesized factor structure fits observed data (confirmatory FA)

Difference between FA and PCA

FA and PCA have similar themes, i.e., to explain covariation between variables via linear combinations of other variables.
However, there are distinctions between the two approaches:

- FA assumes a statistical model that describes covariation in observed variables via linear combinations of latent variables
- PCA finds uncorrelated linear combinations of observed variables that explain maximal variance (no latent variables here)

FA refers to a statistical model, whereas PCA refers to the eigenvalue decomposition of a covariance (or correlation) matrix.

Factor Model

- Orthogonal factor model:

$$\mathbf{x}_{p\times 1} = \boldsymbol{\mu} + \mathbf{L}_{p\times m}\mathbf{f}_{m\times 1} + \boldsymbol{\varepsilon}. \tag{2.1}$$

- The elements $F_1, ..., F_m$ of $\mathbf{f}$ are r.v.s called *common factors*;
- $\mathbf{L}$ is a non-random matrix of *factor loadings* with rank $m \leq p$.
- The mean-zero random errors (specific factors) ε_i are uncorrelated with each other and also with the F_j;
- we assume as well that

$$\begin{aligned} E[\mathbf{f}] &= \mathbf{0},\ \text{COV}[\mathbf{f}] = \mathbf{I}_m, \\ \text{COV}[\boldsymbol{\varepsilon}] &= \boldsymbol{\Psi} = diag(\psi_1, ..., \psi_p). \end{aligned}$$

- That $\text{COV}[\mathbf{f}] = \mathbf{I}_m$ motivates the name orthogonal factors. The diagonal elements of $\boldsymbol{\Psi}$ are termed uniquenesses or specific variances.

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Factor Model

- The structure implies that

$$\text{COV}\left[\mathbf{x}\right] = \mathbf{L}\mathbf{L}^T + \mathbf{\Psi} \text{ and } \text{COV}\left[\mathbf{x}, \mathbf{f}\right] = \mathbf{L}.$$

- Then $\text{COV}[X_i, F_j] = l_{ij}$ — the loading of the ith variable on the jth factor
- and

$$\text{VAR}\left[X_i\right] = \sigma_{ii} = h_i^2 + \psi_i,$$

where the ith communality — the part of the variance arising from the factor loadings – is

$$h_i^2 \stackrel{def}{=} \left(\mathbf{L}\mathbf{L}^T\right)_{ii}.$$

- A simple example is $p = 3$, one factor ($m = 1$), $\mathbf{L} = \mathbf{1}_p$,

$$\mathbf{\Sigma} = \text{COV}\left[\mathbf{x}\right] = \begin{pmatrix} 1+\psi_1 & 1 & 1 \\ 1 & 1+\psi_2 & 1 \\ 1 & 1 & 1+\psi_3 \end{pmatrix}.$$

Factor Model

- Of course not every covariance matrix $\mathbf{\Sigma}$ can be written as $\mathbf{L}\mathbf{L}^T + \mathbf{\Psi}$ with $\mathbf{\Psi}$ a diagonal variance matrix (if $m < p$ is given).
- The model is scale invariant in that if $\mathbf{x}$ follows model (2.1) then so does $\mathbf{y} = \mathbf{C}\mathbf{x}$ with $\mathbf{C}$ diagonal (required so that $\text{COV}[\mathbf{C}\boldsymbol{\varepsilon}] = \mathbf{C}\mathbf{\Psi}\mathbf{C}^T$ is diagonal).
- There is an identifiability problem: given an $m \times m$ orthogonal matrix $\mathbf{\Gamma}$, we can replace $\mathbf{L}$ by $\mathbf{L}\mathbf{\Gamma}$ and $\mathbf{f}$ by $\mathbf{\Gamma}^T\mathbf{f}$ without altering the structure of $\mathbf{\Sigma}$.
- One typically rotates the matrix $\mathbf{L}$, choosing a particular version $\mathbf{L}\mathbf{\Gamma}$ according to an ease of interpretation criterion.

Principal Component Solution

- One (approximate) factorization of the covariance matrix starts by representing it as

$$\mathbf{\Sigma} = \mathbf{\Gamma}\mathbf{\Lambda}\mathbf{\Gamma}^T = \sum_{j=1}^{p} \lambda_j \gamma_j \gamma_j^T,$$

- where $\lambda_1 \geq \cdots \geq \lambda_p$ are the eigenvalues and $\{\gamma_j\}$ the eigenvectors, and approximating this as

$$\mathbf{\Sigma} \approx \sum_{j=1}^{m} \lambda_j \gamma_j \gamma_j^T = \mathbf{L}\mathbf{L}^T, \text{ for } \mathbf{L} = \left(\sqrt{\lambda_1}\gamma_1 \vdots \cdots \vdots \sqrt{\lambda_m}\gamma_m\right).$$

- Then the approximation is improved by adding in $\mathbf{\Psi}$, with diagonal elements $\psi_i = \sigma_{ii} - \left(\mathbf{L}\mathbf{L}^T\right)_{ii}$, i.e.

$$\mathbf{\Psi} = diag\left(\mathbf{\Sigma} - \mathbf{L}\mathbf{L}^T\right).$$

Principal Component Solution

- Given p-dimensional data $\mathbf{x}_1, \ldots, \mathbf{x}_n \sim (\boldsymbol{\mu}, \boldsymbol{\Sigma})$ this is applied to the covariance matrix $\mathbf{S}_{p\times p}$ (or the correlation matrix $\mathbf{R}$), yielding estimates $\hat{\mathbf{L}}$ and $\hat{\boldsymbol{\Psi}}$ and a residual matrix

$$\mathbf{S} - \left(\hat{\mathbf{L}}\hat{\mathbf{L}}^T + \hat{\boldsymbol{\Psi}}\right).$$

- Using the Frobenius norm $\|\mathbf{A}\|^2 = tr\mathbf{A}^T\mathbf{A} = \sum_{ij} a_{ij}^2 \; (= \|vec\mathbf{A}\|^2)$, we have

$$\left\|\mathbf{S} - \left(\hat{\mathbf{L}}\hat{\mathbf{L}}^T + \hat{\boldsymbol{\Psi}}\right)\right\|^2 \leq \left\|\mathbf{S} - \hat{\mathbf{L}}\hat{\mathbf{L}}^T\right\|^2 \leq \hat{\lambda}_{m+1}^2 + \cdots + \hat{\lambda}_p^2; \qquad (3.2)$$

a guide to the choice of m.

Principal Component Solution

- The contribution of the jth common factor towards the variance s_{ii} is $\hat{l}_{ij}^2$ (how?) Recall that $s_{ii} = \left(\hat{\boldsymbol{L}}\hat{\boldsymbol{L}}^T\right)_{ii}^2 + \hat{\psi}_i$.
- and so its (proportional) contribution towards the total variance is

$$\frac{\sum_{i=1}^p \hat{l}_{ij}^2}{tr\mathbf{S}} = \frac{\left\|\sqrt{\hat{\lambda}_j}\hat{\gamma}_j\right\|^2}{tr\mathbf{S}} = \frac{\hat{\lambda}_j}{tr\mathbf{S}} = \frac{\hat{\lambda}_j}{\sum_{j=1}^p \hat{\lambda}_j}; \tag{3.3}$$

  a better guide to the choice of m.
- In R, we use `princomp(X, cor=TRUE)` to start it off.

Maximum Likelihood Solution

- Here one assumes normality, and maximizes the likelihood over $\boldsymbol{\mu}$ (obtaining the mle $\bar{\mathbf{x}}$) and $\boldsymbol{\Sigma}$, subject to $\boldsymbol{\Sigma} = \mathbf{L}\mathbf{L}^T + \boldsymbol{\Psi}$.
- That is to replace

$$L(\boldsymbol{x}; \boldsymbol{\mu}, \boldsymbol{\Sigma}) = \sum_{i=1}^{n} (2\pi)^{-p/2} |\boldsymbol{\Sigma}|^{-1/2} \exp\left\{ -\frac{(\mathbf{x}_i - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\mathbf{x}_i - \boldsymbol{\mu})}{2} \right\}$$

by

$$L(\boldsymbol{x}; \boldsymbol{\mu}, \boldsymbol{L}, \boldsymbol{\Psi}) = \sum_{i=1}^{n} (2\pi)^{-p/2} |\mathbf{L}\mathbf{L}^T + \boldsymbol{\Psi}|^{-1/2} \exp\left\{ -\frac{(\mathbf{x}_i - \boldsymbol{\mu})^T (\mathbf{L}\mathbf{L}^T + \boldsymbol{\Psi})^{-1} (\mathbf{x}_i - \boldsymbol{\mu})}{2} \right\}$$

- Because of the lack of identifiability a further constraint must be imposed; say $\mathbf{L}^T \boldsymbol{\Psi}^{-1} \mathbf{L} = \boldsymbol{\Delta} = diag\,(\delta_1 \geq \cdots \geq \delta_m)$.
- Under this constraint $\mathbf{L}$ is uniquely defined (up to multiplication of its columns by ± 1) if as well the δ's are distinct.

Maximum Likelihood Solution

- First note that if $\mathbf{L}$ is not uniquely defined, then there are $\mathbf{L}_{p\times m}$ and $\mathbf{K}_{p\times m}$ with $\mathbf{L}\mathbf{L}^T = \mathbf{K}\mathbf{K}^T$.
- Then necessarily

$$\mathbf{K} = \mathbf{L}\boldsymbol{\Gamma} \text{ for some orthogonal } \boldsymbol{\Gamma}.$$

- By the singular value decomposition (proof to follow) we can write $\mathbf{L} = \mathbf{G}\mathbf{D}\boldsymbol{\Gamma}_L$, where $\mathbf{G}_{p\times p}$ is the orthogonal matrix of eigenvectors of $\mathbf{L}\mathbf{L}^T$ hence of $\mathbf{K}\mathbf{K}^T$),
- $\mathbf{D}_{p\times m}$ has the singular values of $\mathbf{L}$ — the roots of the nonzero eigenvalues of $\mathbf{L}\mathbf{L}^T$ (hence of $\mathbf{K}\mathbf{K}^T$) — on its diagonal and zeros elsewhere, and $\boldsymbol{\Gamma}_L$ is the orthogonal matrix of eigenvectors of $\mathbf{L}^T\mathbf{L}$.
- Thus we can write $\mathbf{K} = \mathbf{G}\mathbf{D}\boldsymbol{\Gamma}_K = \mathbf{L}\boldsymbol{\Gamma}$, where $\boldsymbol{\Gamma} = \boldsymbol{\Gamma}_L^T\boldsymbol{\Gamma}_K$ is orthogonal.

Maximum Likelihood Solution

- Now, if as well $\mathbf{L}^T\mathbf{\Psi}^{-1}\mathbf{L} = \mathbf{\Delta}$ (*) and $\mathbf{K}^T\mathbf{\Psi}^{-1}\mathbf{K} = \tilde{\mathbf{\Delta}}$, with $\mathbf{\Delta}$ and $\tilde{\mathbf{\Delta}}$ diagonal, then $\tilde{\mathbf{\Delta}} = \mathbf{\Gamma}^T\mathbf{\Delta}\mathbf{\Gamma}$.
- Thus $\mathbf{\Delta}$ and $\tilde{\mathbf{\Delta}}$ have the same eigenvalues – i.e. the same diagonal elements.
- If these diagonal elements are ordered as described then $\mathbf{\Delta} = \tilde{\mathbf{\Delta}}$ and so $\mathbf{\Gamma}\mathbf{\Delta} = \mathbf{\Delta}\mathbf{\Gamma}$.
- Comparing the $(i,j)^{th}$ elements gives $\gamma_{ij} = 0$ if $i \neq j$, hence $\mathbf{\Gamma}$ is diagonal with diagonal elements ± 1.
- One can replace $\mathbf{\Psi}$ in (*) by any invertible matrix; another common choice is $diag\,(\mathbf{\Sigma})$.

Maximum Likelihood Solution

- The R function `factanal()` computes the mle on the basis of the sample correlation matrix $\mathbf{R}$.

Summary and Remark

- Factor Analysis
- PCA & MLE